



Customer Churn Prediction using Deep Learning

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Introduction

- **Objective:** Predict and prevent customer churn in the telecom industry.
- **Importance:** Churn impacts profitability and customer satisfaction.
- **Dataset:** "Telco Customer Churn" dataset from Kaggle.

LITERATURE REVIEW

- Chollet, F., et al., 2015 - "Keras": Keras is an open-source deep learning library that provides a high-level interface to build and train neural networks. It simplifies the process of constructing deep learning models, making it accessible for practitioners. Keras is widely used for customer churn prediction.
- Guyon, I., & Elisseeff, A., 2003 - "An Introduction to Variable and Feature Selection": This paper discusses the importance of feature selection in machine learning. Feature selection methods help identify the most relevant variables, reducing model complexity and improving model performance.
- Bergstra, J., & Bengio, Y., 2012 - "Random Search for Hyper-Parameter Optimization": Random search is a simple but effective technique for hyperparameter tuning. This paper demonstrates the benefits of random search over grid search in finding optimal hyperparameters.
- Smith, L. N., 2018 - "A disciplined approach to neural network hyper-parameters: Part 1 -- learning rate, batch size, momentum, and weight decay": This work provides practical guidance on tuning hyperparameters like learning rate, batch size, momentum, and weight decay, which are critical for training deep learning models effectively.

Project Objectives

1. Data Exploration and Preprocessing

1. Understand the dataset and features.
2. Handle missing values.
3. Encode categorical variables.
4. Scale numerical features.

2. Feature Engineering

1. Identify relevant features.
2. Create new features for predictive power.

3. Deep Learning Model

1. Build a deep learning model using TensorFlow/Keras.

3. Deep Learning Model (Cont.)

2. Experiment with different architectures.

4. Model Training and Evaluation

1. Train the model on the dataset.
2. Evaluate performance using metrics.

5. Hyperparameter Tuning

1. Optimize model performance.
2. Tune learning rates, batch sizes, and activation functions.

6. Interpretability

1. Analyze feature importance.

METHODOLOGY

- Data Preprocessing

Cleaned data by addressing missing values and encoding categorical variables.

Performed feature scaling for consistent input to the neural network.

- Feature Engineering

Identified relevant features including customer demographics and usage patterns.

Created new features like usage ratios and tenure-based features.

- Model Building

Implemented a deep learning model using Keras.

Experimented with different architectures.

- Training and Evaluation

Split the dataset into training and testing sets.

Addressed overfitting through techniques like early stopping and dropout layers.

- Hyperparameter Tuning

Performed hyperparameter tuning to find optimal settings.

- Interpretability

Analyzed feature importance using techniques like SHAP values or feature gradients.

Progress Review

- Data preprocessing, including missing value handling and encoding, completed.
- Feature engineering, including the creation of new features, conducted.
- Default deep learning model trained and evaluated.
- Default hyperparameters used, with successful model training.
- Interpretability explored, although some issues encountered.

Feature importance using Random Forest Model

- Tenure, number of months customer has been with the company.
 - Contract, monthly, yearly, or multiyear.
 - Total Charges
 - Monthly Charges
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- All those features are consistently identified as important features by both metrics.

THANK YOU

